

Build Status — What Is Done and What Is Not

As of 19 September 2026, end of session. Counts in this document are generated from the repository, not maintained by hand: the block results come from `sim/run_sim.py`, the unknowns from the specification's own register.

At a glance

	Count
Circuit blocks simulated	24
Blocks passing every check	17
Blocks failing a check	7 — see below; five of them are new <i>information</i> , not new faults
Research memos	12
Documents in the bible	18
Unknowns closed	4
Unknowns partly answered	4
Unknowns still open	9
Phase	1.5 — block simulation, nearly complete

The pass rate went down on 19 September, and that is progress. It was 22 of 24 that morning. Nothing broke: the temperature-corner pass found six blocks that fail cold or hot and recorded all six in RESULT NOTE prose, where the suite could not see them. They are now enforced checks. A design that was always marginal at $-40\text{ }^{\circ}\text{C}$ now says so when you run it.

Nothing has been captured in KiCad, no PCB exists, and no firmware has been written. Everything below describes design work verified by simulation, which is the stage before schematic capture.

1. Circuit blocks — simulated and passing

Each block is a SPICE netlist plus a set of falsifiable checks. A block "passes" when every claim it makes about itself holds. These are design verification, not layout.

1.1 Power chain — 9 blocks

Block	What it covers	Status
emi_filter	Two-stage LC, CISPR 25 conducted, differential mode	Passing
reverse_battery	Ideal-diode P-FET vs series Schottky	Passing
transient_clamp	ISO 7637-2 pulse 2a, +112 V / 2 Ω / 50 μ s	Passing
load_dump	ISO 7637-2 pulse 5b, 40 V / 0.5 Ω / 400 ms	FAILS — deliberately
buck_preregulator	6–40 V to 5 V at 400 kHz	Passing
sensor_rail	5 V sensor distribution, one PTC per group	FAILS — cold corner
mcu_pdn	3V3 decoupling impedance, 100 m Ω target	FAILS — cold corner
supervisor	Watchdog, brownout, fail-safe gate kill path	Passing
negative_pulses	ISO 7637-2 pulses 1 and 3a, negative excursions	FAILS — real gap

Why **load_dump** fails, and why that is correct. The TVS chosen against pulse 2a absorbs 46 J at 116 W under pulse 5b — roughly 100 $\times$ both its single-pulse energy rating and its continuous dissipation rating. The failure is the deliverable: it says the part selection is wrong, not that the simulation is. The fix is a higher standoff voltage so the TVS stays off during a normal clamped load dump, rather than a larger part to absorb it. **Blocked on U16** — whether the alternator is suppressed at all decides which pulse applies.

1.2 Sensor front-ends — 7 blocks

Block	What it covers	Status
sensor_ratiometric	Pins 41/35/80/37, 0.5–4.5 V into a 3.3 V ADC	Passing
sensor_differential	Shared sensor ground on pin 34, single-ended vs differential	Passing
ntc_frontend	Pin 79 charge-air temperature, divider vs current source	Passing
battery_sense	Pins 04/06, 6–40 V rail into a 3.3 V ADC	Passing
discrete_input	Pins 20/24/71, active-high switched battery	Passing
trip_module_sense	Trip-module state, supervised three-state loop	Passing
cam_frontend	Pin 46 cam Hall, 5 V swing divided into the 3.3 V domain	Passing

ntc_frontend simulates **both** candidate topologies because **U2** is unresolved. One resistance measurement at the machine picks between them; the front-end is designed so that measurement selects a populate option rather than unblocking a redesign.

1.3 Speed and position — 1 block

Block	What it covers	Status
vr_conditioner	Crank VR sensor, pins 52/74/30	Passing

1.4 Injection — 3 blocks

Block	What it covers	Status
injector_boost	Why the boost stage exists, pins 03/05	FAILS — both corners
injector_turnoff	Turn-off recirculation into the boost rail	Passing
boost_converter	Injector rail reservoir sizing	Passing

`boost_converter` passes but carries a recorded gap: it has **no bleed path**. Its charge source regulates the rail from below only. Harmless while current flowed outward; `injector_turnoff` now pushes energy back in, and a hold-current cutoff would climb the rail about 2 V per event with no way down.

1.5 Actuators — 3 blocks

Block	What it covers	Status
metering_unit_pwm	Pin 88, low-side PWM into the fuel metering solenoid	FAILS — both corners
relay_driver	Pins 50/69, low-side FET with flyback	Passing
egr_hbridge	Pins 59/81, positional actuator, feedback on 37	Passing

1.6 Communications — 1 block

Block	What it covers	Status
can_termination	Split vs single termination, J1939 250 kbit/s	FAILS — tempco

1.7 The seven failures, and what each one means

A failing block here is a finding, not a broken simulation. Three different kinds:

Wrong part selected — 1 block.

- `load_dump` — the TVS absorbs 38 J under pulse 5b against roughly a 0.5 J single-pulse class rating. The fix is a *higher standoff* voltage so the part stays off during a normal clamped dump, not a bigger part to absorb it. Blocked on U16: if the alternator turns out unsuppressed, the applicable pulse is 5a and the topology changes rather than the part number.

Missing component — 1 block.

- `negative_pulses` — the buck's VIN pin is rated **−0.3 V to 100 V**, and the negative limit is not a mirror of the positive one. It sees −40.7 V on pulse 1 and −27.8 V on pulse 3a. The TVS clamps correctly; −38 V is a good clamp that is still two orders of magnitude outside what the buck tolerates. Nothing in the chain addresses a negative excursion at all.

Marginal at a temperature corner — 5 blocks. These were always true; until 19 September nothing asked.

- **sensor_rail** — 4.148 A against a real 3.0 A polyfuse trip ceiling, at the stacked cold-and-tolerance corner.
- **mcu_pdn** — 196.7 mΩ against the derived 100 mΩ target at a 4× cold-ESR derate. The bulk electrolytic's ESR *is* the damping in this block, so cold attacks the mechanism the design depends on.
- **can_termination** — 123.96 Ω differential against J1939's 120 ± 2, from 200 ppm/°C resistor tempco alone. An ordinary part, not an exotic one.
- **metering_unit_pwm** and **injector_boost** — their *target* checks fail while their *control-law invariants* hold. Both halves are checked deliberately so a reader sees which is fragile. A target that moves with temperature is not the same as a design that stops working.

One block passes in a way worth reading. **emi_filter** was not failed at its corner, because its −40 dB gate is documented in its own final check as being of unknown adequacy pending measured hardware emissions. Failing a corner against an admittedly arbitrary number manufactures a verdict. The X7R −15% corner is recorded as regression pins with a note that the value sits below the nominal gate.

2. Circuits not yet built at all

These have no netlist, no schematic and no checks. They are the real gaps.

Missing circuit	Pins	Why it is not built
Negative-input protection for the buck	pin 21 chain	Found 19 Sep. The LM5164's VIN is rated −0.3 V to 100 V; the negative limit is not a mirror of the positive one and nothing was sized against it. The TVS clamps correctly to −38 V, which is still two orders of magnitude outside what the buck tolerates. A missing component, not a mis-sized one
Intake throttle driver	6-pin connector	Blocked on U8. Reported battery-fed, which answers who supplies it, not who commands it. May be out of scope entirely
Pre/post-heat output	unknown	Blocked on U14. Same distinction — battery-fed is not the same as ECU-switched. The DSE4522 enables both at 50 °C and this design has no heater output

3. Verification not yet done

The blocks above are each simulated in isolation, at one temperature, against the stimuli named. These are the gaps in the *verification*, not in the circuits.

Gap	Consequence
No temperature corner anywhere	Done 19 Sep, −40/+125 °C across all blocks. Six fail at a corner — emi_filter and can_termination from <i>ordinary specified</i> component behaviour, not stacked worst cases. Being converted from prose into enforced checks

Gap	Consequence
ISO 7637-2 negative pulses unsimulated	Done 19 Sep. The TVS is now modelled as the bidirectional part it is, which is what made the question answerable. Result: the buck's VIN sees -40.7 V against a sourced -0.3 V absolute maximum — see the new gap below
Common-mode EMI unmodelled	<code>emi_filter</code> covers differential mode only. Most real automotive failures are common mode, and modelling it honestly needs a layout to give the return path
Reverse-battery transient — now load-bearing	The switch model turns off instantly, so it always looks perfect. That was a caveat; since <code>negative_pulses</code> it is a design dependency. If the controller opens fast enough the buck is isolated; if not it sees the full clamped negative voltage. No controller part is chosen and no turn-off delay is sourced, so both bookends are simulated and neither is claimed
Watchdog fault-detection time	<code>supervisor</code> bounds only what happens <i>after</i> reset asserts, not how long detection takes

4. Documents

Document	State
Reconciled Specification	Binding. Rewritten §3 on the fitted controller, §4 gained a hardware fail-safe requirement, register extended to U16
Field Handbook	Current
Sensor Reference	Current
Circuit Reference	Generated from the simulations at build time, so it cannot drift
Memos 01-12	Dated records. Superseded conclusions carry banners rather than being rewritten

Memo 01 and Memo 02 are substantially wrong and say so at the top.

Memo 01 misidentified the engine and advised discarding the correct search term. Memo 02 analysed a controller that is not fitted. They are kept because the reasoning that produced them is the record of how the error happened.

5. Unknowns

Closed — 4

#	Question	Answer
U11	Engine identity	36K550ETA 4SR1 , app code 6K3.8703 , CPCB IV+ under certificate P25/2825/24
U1	ECU connector	Bosch 1 928 405 192 / 194 , code C keying
U13	Which controller is fitted	DSE4522 MKII AMF, not the KG640C the diagram shows
U12	Does the fuel relay remove power?	No — it only signals

Three of these four closed against what this project had reasoned. That is the single most useful thing to know about the desk work: it is productive, and it is not reliable without the machine.

Partly answered — 4

U4 CAN expectations (addresses 234 and 44 confirmed) · U5 injector drive profile (family envelope only, no part number) · U8 throttle (supply known, control not) · U14 pre/post-heat (supply known, control not)

Open — 8

U2 charge-air sensor topology · U6 pin 09 function · U7 pins 04/06 sense or feed · U9 catalyst sensor routing · U10 SENT signal identity · U15 what CRS-878 denotes · U16 alternator rectifier suppressed or not · plus U12's contact-type confirmation, which gates nothing

6. Decisions open, and blocking

Decision	Blocks	Why it cannot wait
Integrated driver IC vs discrete FETs	—	DECIDED 19 Sep 2026, memo 12. Discrete FETs, keeping the 100 V rail. MC33816 caps at 72 V, L9781 at a reported 80 V, and both are pre-drivers that need external FETs anyway
Passive pulldown vs active clamp	—	DECIDED 19 Sep 2026, memo 12. Active clamp, built discretely, at every gate. The integrated route relocates this problem rather than removing it
MC33816 last-time-buy 06/08/2027	Fallback path only	Needs a distributor conversation, not another search
L9781 datasheet unretrievable	Confirming the decision's runner-up	Five failed fetches across two sessions. Needs FAE or distributor access
Air-intake shutoff	Nothing electrical	Owners' decision. Nothing in this design can stop an oil-fuelled runaway
Trip-module pickup target	Module ordering	Whether it shares our flywheel target. Sharing is cheaper; separate preserves the independence that is the reason for fitting it

7. Phases

Phase	Output	State
0	Reconciled specification	Done
1	Research memos	Done — twelve
1.5	Block-level simulation	In progress — 23 blocks, 22 passing
2	KiCad hierarchical schematic capture	Not started
3	4-layer PCB layout, DRC, fab outputs	Not started
4	Firmware skeleton	Not started

What ends phase 1.5: the four missing circuits in §2, the verification gaps in §3, and the verification gaps in §3. The two driver decisions in §6 are now taken. What starts phase 2: assigning PTxx pins to the 94-way connector — which is also what unblocks the per-pin detail of the fail-safe requirement.

8. Work at the machine

Ordered by what it unblocks.

Task	Closes	Unblocks
Alternator rectifier part number, or a scope during a live disconnect under load	U16	The whole input-protection topology, and load_dump 's failure
Resistance across the charge-air element, cold	U2	The ntc_frontend populate choice
Follow the throttle's 6-pin cable to where it ends	U8	Whether a throttle driver exists at all
Trace who switches the heater feed	U14	Whether a heater output is in scope
Injector part number from the body	U5	The boost rail's target voltage
Relay contact-type check	U12 confirmation	Nothing — recorded for completeness

9. Resume here — next session

Written at the end of 19 September so the next session does not have to re-derive where things stand. Nothing is half-finished: the tree is clean, the suite passes, everything is pushed.

Done since this section was written

Both items below are complete: the TVS is bidirectional, ISO 7637-2's negative pulses are simulated, and temperature corners are run and enforced across all 24 blocks. What they produced is in §1.7 and §2. The text of the two items is kept as written because it records what was expected, and the negative-pulse result was not it.

The two pieces of phase 1.5 that were outstanding — now done

1. Replace the TVS model, then simulate the negative pulses. In that order — the second is not answerable until the first is done.

sim/blocks/transient_clamp.cir and load_dump.cir share this model:

```
.model TVS D(Is=1e-12 N=1.6 Rs=0.35 BV=36.7 IBV=1e-3)
```

That is a **single diode**. The real part is an SMBJ33CA, and the `CA` suffix means bidirectional. A single diode conducts forward at 0.7 V in reverse polarity, so any reverse-polarity or negative-pulse question put to it gets a confident wrong answer rather than a visible failure.

Build a bidirectional subcircuit — two junctions back to back — and switch both blocks to it. **Both must reproduce their current results:** `transient_clamp` passing, `load_dump` failing at ~46 J absorbed. If either moves, that is a finding about the old model and should be reported, not tuned away.

Then add a `negative_pulses` block for ISO 7637-2 pulses 1 and 3a on a 12 V system. Spec §6 requires ISO 7637-2 and only 2a and 5b exist. Take severity from memo 06 if it is there; otherwise state the values and label them INFERRED.

2. Temperature corners across all 23 blocks. Every result in this repo is a 27 °C result. The tolerance pass built the pattern to follow — see `load_dump.cir` and `supervisor.cir` for how a corner is stated. This is the piece most likely to be cut off part-way, and it survives that well: each block's corner is self-contained, which is how the tolerance pass came back cleanly from being truncated at 11 of 19 blocks.

Also outstanding

Four artifacts built on 19 September have never been independently reviewed: `egr_hbridge` , `trip_module_sense` , `cam_frontend` , and `docs/pinmap.md` . The last review ran before any of them existed. Both review passes so far found real errors, including a 2× arithmetic slip inside the block written to fix a 1000× one.

Needs a person, not an agent

Item	Why
MC33816AE last-time-buy 06/08/2027	Distributor conversation. Gates memo 12's fallback path
L9781 datasheet	Five failed retrievals across two sessions. Needs FAE or distributor access. It is the runner-up to the driver decision and cannot be confirmed without it

At the machine

U16 is now the one that gates most — whether the alternator's rectifier is suppressed decides which ISO 7637-2 pulse applies and therefore whether `load_dump` 's failure is fixed by a part number or a different circuit. Then U2, U8, U14, U5.

The pattern worth carrying forward

Across 19 September, four wrong numbers were found in prose — comments, memos, the specification — and none in a check. A factor of 1000, a factor of 6, a factor of 2, and a stale 0.19 V that should have been

0.308 V. Every one of them became or nearly became a requirement, and every one was caught by something executing the claim rather than reading it.

Two circuits have now been specified in a way that would have destroyed an MCU pin. The first was caught by a simulation block. The second — the cam input — survived precisely because no block existed for it.